UCC602: ENVIRONMENTAL ENGINEERING

L	T	P	Cr
3	**1**	**2**	**4.5**

Course Objective: To introduce the water supply and sanitation systems, designing the components and suitable treatment processes associated with the water supply and sanitation systems. This course will also provide the exposure of software to design the water supply distribution system

Water and water supply system: Source of water, impurities in water and their effect, domestic water quality standards; water demand and quantity estimation, Intakes, distribution of water, and design of distribution system.
Water treatment*:* Water treatment plants and components; Sequencing of unit operations and processes, coagulation-flocculation, slow sand and rapid gravity filtration, disinfection, softening, ion exchange and adsorption.
Wastewater system: Quantification of sewage; Characterization of sewage; Types of sewerage systems; Hydraulic design of sewers, sewer outfalls and sewer appurtenances
Waste water treatment: Components of domestic wastewater treatment, Screening, Grit removal, sedimentation, biological processes, microbial growth kinetics, aerobic and anaerobic process, nitrification and denitrification, trickling filter, Activated Sludge process, sludge digestion process, Waste stabilization pond systems, Anaerobic stabilization units, UASB reactors, Natural wastewater treatment systems.
Software application: Design of water supply distribution system using Water GEMS/EPANET.

Laboratory work:
pH, acidity, alkalinity and hardness testing; DO, BOD and COD; Solids (TSS, VSS and TDS); Nutrients(TKN, TN and TP); SVI and Settling tests; Chlorination, residual chlorine and MPN test; Oil and grease and pesticides; Iron, fluorides, sulfates, chlorides, sulfides and phenols

Course Learning Outcomes (CLO):
Upon completion of this course, the students will be able to:

1. Characterize water and wastewater
2. Evaluate and design a suitable municipal water treatment unit
3. Evaluate and design a suitable sewerage treatment unit

Text books:

1. *Garg,S.K,EnvironmentalEngineering,Vol.I,.KhannaPublishers,New-Delhi.*
2. *CPHEEO ManualonWaterSupplyandTreatmentbyMinistryofUrbanDevelopment,NewDelhi.*
3. *CPHEEO Manualonsewerageandsewagetreatment,MinistryofUrbanDevelopment,NewDelhi.*

4. *P.N.* Modi*;SewageTreatmentanddisposal&WasteWaterEngineering,StandardBook House New-Delhi.*

Referencebooks:

1. *Metcalf Eddy, WastewaterEngineering,McGrawHill.*
2. *Peavy,RoweandTchobanglous,EnvironmentalEngineering,McGraw Hill.*
3. *Fair,Geyer&Okun, WaterandWasteWaterEngineering (Vol.1 & 2),JohnWiley,NewYork.*
4. *Sawyer,McCarty&Parkins,EnvironmentalChemistry,McGrawHill.*
5. *StandardMethodsfortheExaminationofWaterandWasteWater,AmericanPublicHealthAssociation.*